

Semester	B.E. Semester VII – Computer Engineering
Subject	R Programming
Subject Professor In- charge	Prof. Sanjeev Dwivedi
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Laboratory	M-411B

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Grade and Subject Teacher's Signature	

Experiment Number	04
Experiment Title	R Project Implementation and Version Control Using Github

Resources / Apparatus Required	Hardware: Computer system	Software: Colab
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Objective	The objective of this experiment is to implement an image recognition and classification system in R using the Keras deep learning framework. The project aims to demonstrate how a neural network can be trained to classify images into two distinct categories by preprocessing raw image data, building a fully connected neural network architecture, and evaluating its performance on unseen data.
Description	This experiment implements a binary image classification pipeline using R, EBImage, and Keras with a TensorFlow backend. Setup: The EBImage package (Bioconductor) was used for image reading and preprocessing, while keras (with <code>install_keras()</code>) was used to build and train the neural network on a Python/TensorFlow backend running in the Colab environment. Data: The dataset consists of 12 images belonging to two classes (labelled 'p' and 'c'), 6 images per class, uploaded directly into the Colab session. Key operations performed: • Reading images using <code>readImage()</code> from EBImage • Resizing/normalizing images to a fixed dimension so all inputs are consistent in shape • Reshaping image arrays into a flattened matrix format suitable as input to a dense neural network • Splitting data into training and test sets • One-hot encoding class labels using <code>to_categorical()</code> • Building a <code>keras_model_sequential()</code> with dense layers (256 → 128 → 2 units), using activation functions appropriate for a multi-class softmax output • Compiling the model with <code>categorical_crossentropy</code> loss and the RMSprop optimizer • Training the model for 30 epochs with a batch size of 32 and 20% validation split • Evaluating the trained model on the test set and generating predictions • Producing a confusion matrix comparing predicted vs actual labels
Output	<pre>num [1:10, 1:2352] 1 1 1 0.26 0.49 ...</pre>

	<pre>Model: "sequential" Layer (type) Output Shape Param # ===== dense_2 (Dense) (None, 256) 602368 dense_1 (Dense) (None, 128) 32896 dense (Dense) (None, 2) 258 ===== Total params: 635522 (2.42 MB) Trainable params: 635522 (2.42 MB) Non-trainable params: 0 (0.00 Byte)</pre> <pre>loss: 0.217159911990166 accuracy: 0.899999976158142 Actual Predicted 0 1 0 5 1 1 0 4 A matrix: 10 × 4 of type dbl Predicted Actual 0.995167017 0.004833035 0 0 0.995498419 0.004501622 0 0 0.995553911 0.004446057 0 0 0.989129305 0.010870719 0 0 0.986242056 0.013757944 0 0 0.024969654 0.975030303 1 1 0.004608262 0.995391726 1 1 0.013425685 0.986574352 1 1 0.020650603 0.979349315 1 1 0.873651803 0.126348183 0 1</pre>
Conclusion	The implemented model successfully classified the two image categories with a test accuracy of 90%, correctly predicting 9 out of 10 test samples as shown in the confusion matrix. This experiment provided practical exposure to building an end-to-end image classification pipeline in R, covering image preprocessing, neural network design, model training, and evaluation, while also reinforcing debugging skills when resolving package installation issues.